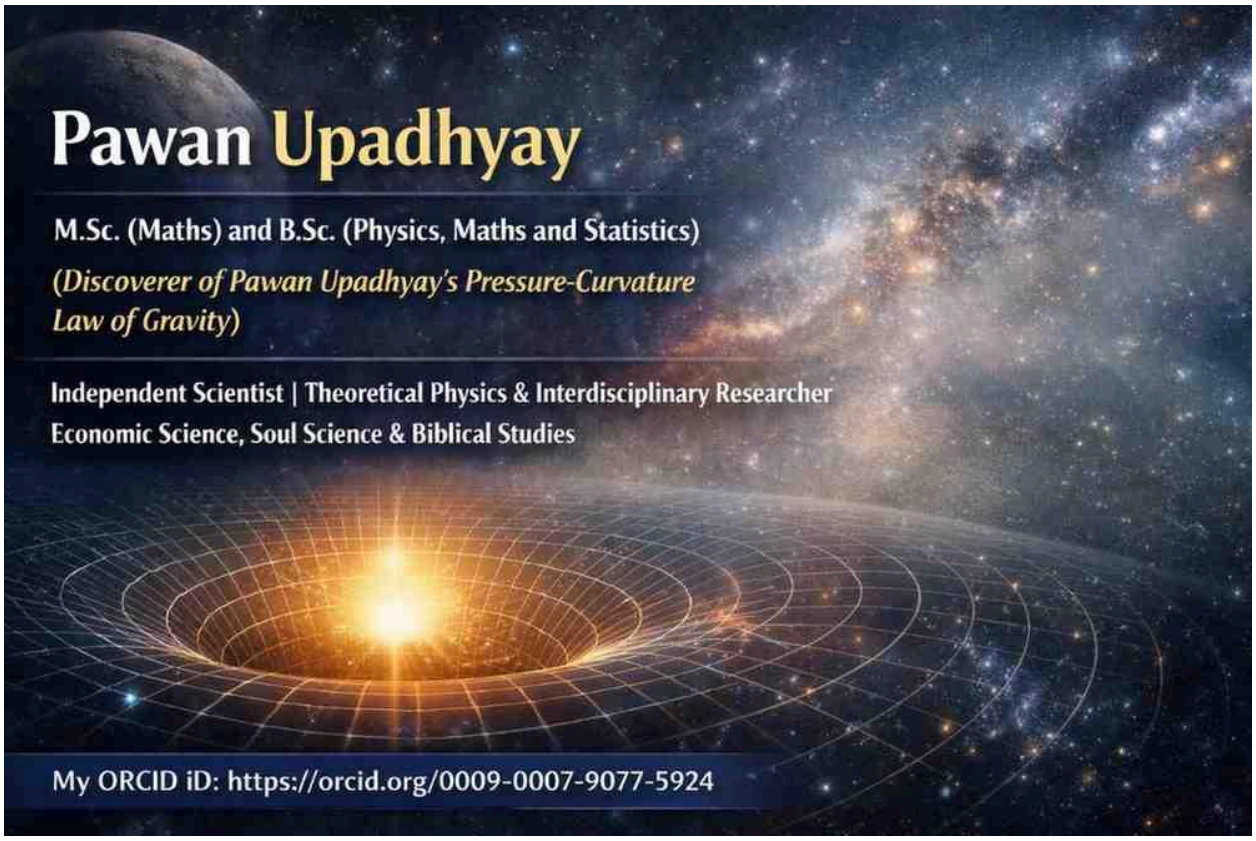
Research Paper: Unit and Dimensional Consistency of My PPC Stress-Energy Tensor with the Einstein Field Equation

Author: Pawan Upadhyay
Affiliation: Independent Researcher and Independent Scientist
Email:pawanupadhyay28@hotmail.com
2nd Email: pawankanyakubj@gmail.com
ORCID iD: https://orcid.org/0009-0007-9077-5924



1. My PPC Stress-Energy Tensor

$T^{ppc}_{uv} = (E_d + P_g)u_u u_v + P_g\, g_{uv}$

2. Einstein Field Equation

$G_{uv} = (8\pi G/c^4)T_{uv}$

3. Substitution of My PPC Stress-Energy Tensor

$G_{uv} = (8\pi G/c^4)[(E_d + P_g)u_u u_v + P_g\, g_{uv}]$

4. Unit of the PPC Stress-Energy Tensor

$[T^{ppc}{}_{uv}] = J/m^3$

5. Unit of the Einstein Tensor

$[G_{uv}] = m^{-2}$

6. Dimensions of Energy Density and Gravitational Pressure

$[E_d] = J/m^3$

$[P_g] = Pa = J/m^3$

7. My Gravitational Pressure Relationship

$P_g = \omega E_d$

$[\omega] = 1$

8. Dimensional Consistency of the Einstein Field Equation

$[G_{uv}] = [(8\pi G/c^4)T^{ppc}{}_{uv}]$

Using SI units:

$[G] = m^3/(kg \cdot s^2)$

$[c] = m/s$

$[T^{ppc}{}_{uv}] = J/m^3 = kg/(m \cdot s^2)$

Therefore:

$[(G/c^4)T^{ppc}_{uv}] = m^{-2}$

Hence:

$[G_{uv}] = m^{-2}$

9. Conclusion

The PPC stress-energy tensor has units of energy density, J/m³. The Einstein tensor has units of inverse square metres, m^{-2}.

The substituted Einstein field equation is dimensionally consistent when all quantities are expressed in SI units and the four-velocity and metric components follow compatible coordinate conventions.

2nd Research Paper

Research Paper: Unit and Dimensional Consistency of My PPC Stress-Energy Tensor with the Einstein Field Equation Using Natural Units (c = 1)

1. My PPC Stress-Energy Tensor

$T^{ppc}_{uv} = (E_d + P_g)u_u u_v + P_g\, g_{uv}$

2. Einstein Field Equation

$G_{uv} = 8\pi G\, T_{uv}$

3. Substitution of My PPC Stress-Energy Tensor

$G_{uv} = 8\pi G[(E_d + P_g)u_u u_v + P_g\, g_{uv}]$

4. Unit of the PPC Stress-Energy Tensor

$[T^{ppc}_{uv}] = L^{-4}$

5. Unit of the Einstein Tensor

$[G_{uv}] = L^{-2}$

6. Dimensions of Energy Density and Gravitational Pressure

$[E_d] = L^{-4}$

$[P_g] = L^{-4}$

7. My Gravitational Pressure Relationship

$P_g = \omega E_d$

$[\omega] = 1$

8. Dimensional Consistency of the Einstein Field Equation

$[G] = L^2$

$[T^{ppc}_{uv}] = L^{-4}$

Therefore:

$[G\ T^{ppc}_{uv}] = L^2 \times L^{-4} = L^{-2}$

Hence:

$[G_{uv}] = [8\pi G\ T^{ppc}_{uv}] = L^{-2}$

9. Conclusion

Using natural units, c = 1:

$[T^{ppc}_{uv}] = L^{-4}$

$[G] = L^2$

$[G_{uv}] = L^{-2}$

The PPC stress-energy tensor and the Einstein field equation are dimensionally consistent when all quantities are expressed in a consistent natural-unit convention.